

Exploring Wearable Design for Emotional Health and Well-Being during Menopause: Perspectives and Design Opportunities

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Abstract

Menopause is a significant and complex life stage characterized by various physical and emotional changes. While recent research has focused on wearable devices for menopause support, most studies have concentrated on monitoring and suppressing physiological symptoms, overlooking emotional challenges. This paper investigates emotional challenges during menopause through an artifact workshop (n=8) in Study 1, and explores design opportunities for supporting emotional health. In Study 2, a focus group (n=6) used "Calmora" as a design probe to examine wearable device functionality and interaction for emotional support. The findings highlight the potential of incorporating natural elements and emphasize the importance of developing non-stigmatizing, encouraging solutions. This study offers design insights for wearable technologies to support emotional health well-being during menopause.

CCS Concepts

• **Human-centered computing** → **User studies**; *Empirical studies in HCI*; *User centered design*.

Keywords

Menopause, Emotional Support, Digital Technologies, Wearable devices

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1 Introduction

Menopause typically occurs between the ages of 45 and 55, although it can last longer [33]. During this period, the vast majority of individuals experience physiological phenomena such as hot flashes and night sweats, which significantly impact their daily lives and work, often accompanied by negative emotional states, including anxiety and depression [39]. Within the HCI community, researchers have started to focus on this phenomenon, aiming to elucidate the experiences of individuals undergoing menopause and exploring how digital technologies can intervene to mitigate these effects.

Among digital technologies, wearable devices have become particularly widespread in the domain of women's health due to their ability to provide seamless monitoring. Over the years, various wearable solutions have been designed and implemented to address a wide range of women's health concerns, including maternal health, menstrual health, fertility, breast health, and menopause [2, 14, 28, 29, 40, 41, 44]. For example, Luo et al. developed smart clothing that helps individuals identify their menopausal stage by monitoring hot flash data, such as frequency, intensity, and other relevant metrics [28]. However, most research has primarily focused on using wearable devices to monitor bodily changes and alert users to these changes. In contrast, there remains a significant gap in exploring how digital technologies can actively alleviate and manage emotional challenges, such as anxiety and unease, that are crucial for menopausal individuals. These emotional aspects are closely linked to hormonal fluctuations and play a vital role in overall well-being.

In the field of HCI, many wearable technology studies focus on alleviating negative emotions, such as using mindfulness meditation techniques to reduce anxiety [3, 10, 52]. Breathing exercises for emotional regulation have been widely proven to be an effective method. However, how to design effective interventions specifically for the menopause context remains an area for further exploration. Research shows that anxiety and unease during menopause are often caused by bodily changes and concerns about aging [34]. These negative experiences often arise unexpectedly, with many

individuals reporting that their temper becomes irritable or that they feel embarrassed due to uncertain symptoms like hot flashes.

To explore the design of wearable devices for emotional intervention and support for individuals undergoing menopause, we conducted two studies. First, in Study 1, we explored which methods could effectively help alleviate the mood of individuals experiencing menopause. Then, in Study 2, based on the results of Study 1, we designed an “Calmora” with natural elements as a probe to uncover insights for designing effective interventions to support the emotional well-being of this group. The findings suggest that design elements that convey “nature” and “vitality” are among the most effective approaches. It is also essential that the design avoid stigmatization and labeling, particularly in its appearance and forms of interaction, to ensure inclusivity and user comfort.

2 Related Work

2.1 Addressing Emotional Challenges During Menopause

Menopause marks the end phase of menstrual cycles that often comes with a combination of physical and emotional challenges, including hot flashes, sleep disturbances, anxiety, mood fluctuations, etc. [7, 32]. Previous studies have demonstrated that medical therapies, especially hormone therapy, can effectively manage menopausal conditions [43, 48]. While the value of medical therapies is undeniable, the needs and experiences of menopause cannot fully be addressed by the current healthcare system, particularly concerning the emotional aspects [23].

Emotional fluctuations during menopause are strongly associated with physical symptoms, such as hot flashes resulting from hormonal fluctuations [46, 47], as well as uncertainties surrounding bodily changes, which can contribute to negative emotional experiences [12]. Breathing exercises and mindfulness have been proven to be effective in previous studies to support emotional well-being [11, 17, 21, 25]. For example, Tsai’s study found that yoga breathing exercises alleviate premenstrual symptoms by reducing physical pain, relieving stress, and improving mental well-being [49].

Furthermore, socio-cultural factors play a critical role in shaping their emotional experiences. Previous studies have highlighted cultural factors that contribute to stigma and hinder a holistic approach to managing menopause. For example, in some places, menopause is often viewed negatively, with gender-specific cultural connotations associated with “madness”, “irritability”, and “mental illness” [24]. Such perceptions not only exacerbate emotional challenges, but also discourage open discussions and holistic management approaches, further isolating women during this critical life stage and highlighting the importance of emotional support.

2.2 Wearables in Menopause Management

Previous studies have explored how wearable technology can be implemented to support people’s emotional well-being. Alhejaili et al. demonstrated that wearable devices can assist users in identifying the sources of their stress and negative emotions, enabling them to develop customized management plans accordingly [1]. Similarly, Jimah et al. conducted a study in which pregnant women used the Oura Ring, a smart ring that monitors physiological and

emotional states. By pairing the ring with a mobile app, participants engaged in data-driven interventions that improved sleep quality and reduced stress, depression, and other mental health issues [19].

Given the complexities of menopause, characterized by uncertainties in physical and emotional changes [23], portable technology has gained prominence for its ability to monitor physiological parameters and offer real-time support for both physical and mental aspects. However, current wearable solutions for menopause focus primarily on monitoring, predicting, and managing physical symptoms. For example, bracelets such as Grace [13] and Embr [16] have been developed specifically for the treatment of hot flashes. These devices provide a sensation on the wrist, which helps in the process of cooling body temperature. Similarly, through a workshop, Tutia et al. introduced the cooling bra idea from one of the participants, which tracks heart rate and skin conductivity to predict hot flashes and provides cooling relief [50]. Beyond physical symptoms management, Griffith proposed the use of a smart ring to monitor and predict depression during perimenopause [18].

Based on these studies, we aim to explore digital technologies that support the reception and management of negative emotions during menopause, particularly those triggered by common symptoms like hot flashes and night sweats. This aspect has not been sufficiently addressed in existing research on emotional experiences of individuals going through menopause.

3 Study One: Understanding Challenges Through the Artifact Workshop

To better understand the emotional changes experienced during menopause and explore possible solutions, Study 1 utilized an artifact workshop approach. This method facilitated participants to share their experiences with negative emotions. During the workshop, participants worked on artifacts to visually represent their feelings and show the solutions they perceived as effective in managing these emotional challenges.

3.1 Method and Data analysis

3.1.1 Participants Recruit. The study recruited 8 participants through community bulletin boards, with an age range of 46 to 66 ($M = 55$, $SD = 6.86$). This sample represents the four stages of menopause: pre-menopause (before symptoms occur), perimenopause (when symptoms start to occur due to hormonal changes), menopause (when menstruation has stopped), and post-menopause (when the last period was over 12 months ago) [26]. The details of the participants are in Table 1 (see appendix). The workshop lasted 2.5 hours and each participant was paid approximately \$ 12 in money.

3.1.2 Workshop Process. During the workshop, participants engaged in discussions about their experiences in managing the negative emotional challenges associated with menopause [12]. Given the cultural context in which discussions on such issues are rare, the workshop used artifacts as scaffolding to facilitate the expression and discussion of participants [8, 24]. The workshop was divided into two parts. In the first part, we introduced the discussion topics by showing videos related to menopause, which encouraged participants to share their experiences and coping strategies. During this process, participants became familiar with each other and built a sense of trust. In the second part, all participants were asked to

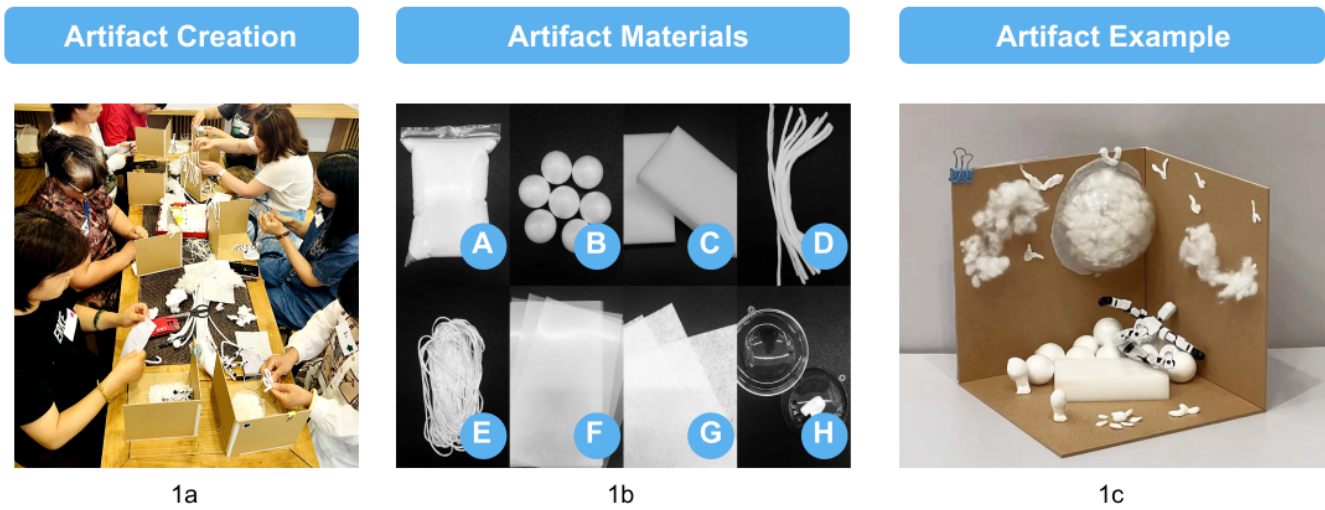


Figure 1: The figure showcases the artifact creation process through three stages. (a) Participants engage in a hands-on workshop, collaboratively assembling artifacts using various materials. (b) A set of materials used in the creation process is displayed, labeled A to H, (A) Cotton stuffing, (B) foam balls, (C) foam sheets, (D) pipe cleaners, (E) white string, (F) plastic sheets, (G) fabric sheets, and (H) a clear container. (c) A completed artifact is presented, demonstrating the use of the provided materials in a creative final design.

use crafting materials (as shown in Figure 1a) to create a method or tool that could help individuals experiencing menopause alleviate negative emotions. Following this, each participant presented their artifact, sharing their proposed solutions and design ideas.

3.1.3 Data Collection and Analysis. The study received university ethics approval. With the consent from participants. Three researchers independently conducted thematic analysis to code the textual data, and two additional researchers reviewed the coding results. Finally, all participants took part in a discussion of the final findings.

3.2 Findings

In this workshop, we did not restrict participants to wearable-based solutions. Instead, our goal was to observe the solutions they proposed through their crafted artifacts and derive design inspirations from these ideas. Based on our data and analysis, we identified three key findings that serve as the foundation for future research and design.

3.2.1 Origins of Negative Emotions. The participants mentioned that physical changes caused by menopause, particularly sudden episodes of hot flashes and night sweats, were the main triggers of their negative emotions. These changes significantly disrupted their daily lives and their work. Many participants noted that *"hot flashes often occur unexpectedly, making me feel out of place in my surroundings"* (P1 workshop) and such sudden changes frequently led to feelings of "embarrassment." Furthermore, uncertainty about these bodily changes or the ineffectiveness of medical treatments often resulted in anxiety and a sense of helplessness, which further intensified their negative emotions. Unlike other forms of anxiety,

these emotions tend to subside as physical symptoms disappear, only to resurface with the frequent recurrence of such symptoms.

3.2.2 Expectations for Non-Medical Treatments. All participants had tried medical treatments, but generally reported unsatisfactory results. As they repeatedly mentioned, sudden onset of physical discomfort during menopause often requires long-term medication to effectively suppress, which is not their preferred solution. Some participants also reported using digital technologies, such as wristbands, during menopause. However, most current devices focus on monitoring and providing reminders, which do not address their core concerns. They emphasized that timely care and emotional support during moments of discomfort are critical. Unfortunately, they noted that such care and attention from family and society is currently minimal.

3.2.3 Natural Elements in Design Solutions. During the presentations of their solution design concepts, we observed that six participants incorporated natural elements as a core strategy to alleviate negative emotions. Among them, five directly utilized concepts of "landscapes and plants," while P6 employed a metaphorical approach, using "clouds" to symbolize a sense of calmness. All participants agreed on the strong healing potential of nature. P5 noted: *"When I imagine myself immersed in nature, I can feel my negative emotions being absorbed by it"*. Therefore, integrating natural elements into the design of products aimed at alleviating emotional distress for menopausal individuals could serve as an insightful approach.

3.2.4 Design Probes Development.

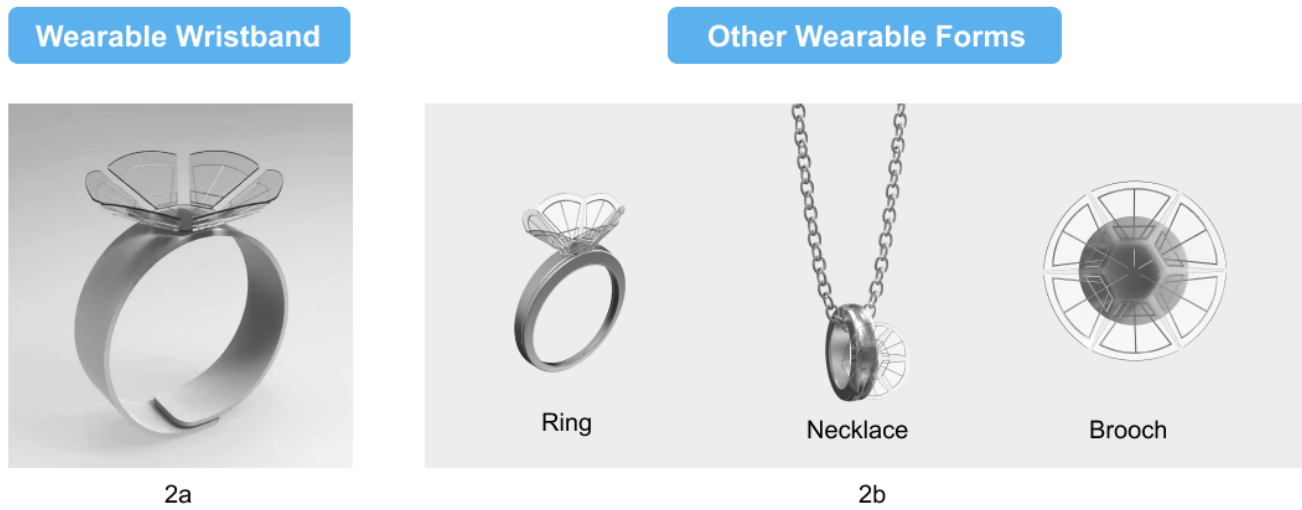


Figure 2: This figure presents wearable design concepts featuring a shared petal-like structure. (a) Displays a wearable wristband with a metallic band and the petal-like structure on top. (b) Showcases three alternative wearable forms—a ring, a necklace, and a brooch—each incorporating the same petal design.

4 Study Two: Investigating Wearable Solutions for Menopause Emotional Support

Through Study 1, we gained a preliminary understanding of the need for timely, non-pharmacological emotional support for menopausal individuals experiencing negative emotions due to drastic physical changes. To further explore design opportunities in this area, we developed a wearable prototype incorporating natural elements based on the findings of Study 1. This prototype was used as a design probe in an online focus group to uncover potential directions for emotional support design in menopausal contexts.

4.1 Method and Data analysis

4.1.1 Designing probes. To address negative emotions triggered by symptoms such as hot flashes, a breathing guidance method can be used to help users regulate their emotions. This approach has been widely validated in mindfulness and meditation research for its positive effects on emotional well-being [31, 35]. Based on this, we designed a wearable wristband, "Calmora", that features a flower-shaped expressive soft robotic component on its top (as shown in figure 2). The device continuously monitors the user's heart rate and body temperature, which have been identified in previous research as key physiological indicators related to hormonal changes and negative emotions [15, 38]. When heart rates or body temperatures reach a predefined threshold, the silicone flower performs petal opening and closing motions, guiding the user through breathing exercises and promoting emotional relief.

In addition to the wristband, we explored other wearable forms, such as rings, necklaces, and brooches, shown in figure 2. These wearables were presented to participants through images and videos as more specific design forms. These prototypes served as design probes, aiming to further stimulate participants' descriptions of potential use scenarios and functionalities. Through their feedback,

we sought to derive design insights better tailored to the needs and experiences of individuals undergoing menopause.

4.1.2 Focus Groups. To further explore design opportunities to alleviate negative emotions during menopause, we used "Calmora" as a design probe and conducted an online focus group. The recruitment was carried out through an online community, with the requirement that the participants were currently experiencing menopause and were proficient in using the Voov online meeting platform. Ethical approval for the study was obtained from the authors' institution. Six participants joined the study 2, with ages ranging from 49 to 55 years ($M = 53.17$, $SD = 1.95$). The details of the participants are in Table 2 (see appendix). The online workshop lasted 1.5 hours and each participant received a compensation of approximately \$ 8.5.

The focus group was divided into three sections. In the first 20 minutes, the participants get familiar with each other by introducing themselves and sharing strategies to manage negative emotions during menopause. In the next 50 minutes, we presented wearable design concepts incorporating natural elements, along with images and videos, as well as existing solutions with different forms and modalities like blankets, handheld devices, and bras. The participants conducted a detailed review of the design concepts, including their aesthetics, functionality, and overall approach. In the last 10 minutes, we collected participants' evaluations of their willingness to adopt these design solutions.

4.2 Findings

4.2.1 Current Strategies. Five participants managed negative emotions during menopause through exercise and social activities, while four used medication but expressed concerns about side effects, preferring alternative solutions. The use of technology, especially wearable devices for psychological support, remains limited. Five

participants used social networks and health apps for information and only one consistently used an Apple Watch to monitor physiological data related to exercise. Most existing technological interventions focus on monitoring and information provision, which aligns with our previous findings.

4.2.2 Positive Perceptions and Proactive Emotional Support. The participants expressed positive attitudes towards our design solution, with an average willingness-to-use score of 4.17 out of 5 ($SD = 0.69$). All participants agreed that they believed that this type of wearable device could offer psychological signals and guidance to help them manage emotional challenges during menopause. F3 highlighted the importance of positive intervention strategies that encourage cognitive changes in the way people perceive and respond to menopause and its symptoms. *"The intervention for menopause should take a positive approach rather than simply focusing on suppression. If symptoms like sweating occur, the goal should not be to stop them directly, but rather to guide individuals on how to manage and cope with them"* (F1 focused group). This perspective underscores how solutions can play a crucial role in fostering a more encouraging and proactive approach to alleviating negative emotions and supporting overall well-being during menopause.

4.2.3 Preferences for Wearable Jewelry and Interaction. Compared to other forms, such as blankets, handheld devices, and bras, all participants preferred the jewelry design, valuing its portability and aesthetic appeal. This preference aligns with their desire for discreet solutions that integrate seamlessly into their daily lives. In addition, participants perceived natural elements, such as flower design, as an effective and meaningful representation that enhances emotional connection and provides a soothing effect.

Regarding the interaction modalities, participants expressed differing opinions. Three participants noted that the opening and closing movements of the petal could increase their anxiety and nervousness. They suggested alternative solutions, such as the incorporation of colors, scents, and audio instructions. Participants also emphasized the importance of flexibility, enabling users to customize interactions to better align with their personal preferences and needs.

5 Discussion

5.1 The Importance of Emotional Support Exceeds Symptom Suppression

Among the technologies discussed with participants, all were designed to directly and temporarily suppress symptoms through external interventions. For example, cooling wristbands such as Grace [13] and Embr [16] primarily rely on physical mechanisms to alleviate symptoms. This raises critical questions about how these wearable solutions differ from traditional pharmacological treatments, which similarly provide only temporary relief without addressing the underlying causes of menopause-related challenges. Such approaches, while effective in the moment, may inadvertently reinforce a reliance on external aids rather than empowering individuals to develop self-management strategies and achieve a mental shift that enables them to navigate menopause with greater confidence.

Participants also questioned the approach of suppressing physiological changes, believing that frequently occurring symptoms like hot flashes and night sweats should not be suppressed, as this may harm the body. What they need is digital technology to support recovery after these symptoms occur and guide them in accepting and managing both the physical changes and the potential negative emotions. Participants expressed a preference for solutions that offer encouragement and guidance on managing their symptoms internally, rather than solely relying on external interventions. In line with the work of Tutia et al. and Cioffi Felice et al. [12, 50], HCI researchers should prioritize "lived experiences", ensuring that technological interventions go beyond symptom suppression to support emotional well-being and foster a sense of control and self-efficacy.

In HCI, there is already a growing recognition of the importance of designing technologies that support self-exploration, emotional awareness, and behavioral change, particularly in the context of women's health [4, 5, 37, 45]. These approaches emphasize equipping individuals with the tools and insights needed to better understand their bodies and emotions, ultimately empowering them to develop sustainable coping strategies. Our design solution, Calmora, reimagines the role of wearable technology in menopause support by shifting the focus from passive symptom relief to active self-empowerment. Participants' feedback on Calmora highlights the potential for wearables to facilitate deeper emotional engagement tailored to menopause challenges, especially for hormonal changes. Moving forward, future research should explore how wearable interventions can be designed to provide proactive and personalized support, enabling users to actively manage their symptoms with confidence and resilience.

5.2 Natural Design Elements and Emotional Support in Menopause

Kellert, in Biophilia Hypothesis, stated that "human dependence on nature ... extends far beyond the simple issues of material and physical sustenance to encompass the human craving for aesthetic, intellectual, cognitive, and even spiritual meaning and satisfaction" [20]. Similar to Ulrich et al.'s stress reduction theory, our findings indicate that natural elements provide "restorative responses" for individuals experiencing menopause, particularly by fostering a sense of calm to help them cope with the unease, especially brought on by hot flashes [51]. Previous studies have further demonstrated that immersing oneself in natural environments can positively impact an individual's health and overall sense of happiness [6, 30, 36].

Many participants have already adopted nature-related strategies, such as hiking, visiting natural parks, cycling, etc. for managing menopausal symptoms. As they navigate menopause, they have expressed a strong desire to connect with nature to counter the effects of aging and regain their energy. A strong bond with nature is closely linked to improved physical and mental well-being, ultimately contributing to a better quality of life [9, 42]. Insights from our focus group confirm participants' preferences and perspectives on incorporating natural elements into wearable devices. Therefore, it is valuable for HCI researchers and designers to explore how nature-inspired elements can be seamlessly integrated into

existing technologies and thoughtfully incorporated into future designs. By harnessing the healing potential of nature and leveraging people's innate biological affinity for it, technology can foster more engaging and appealing ways to support emotional well-being in menopause.

5.3 Subtle Integration of Emotional Support Design for Menopause in Daily Life

Individuals' perceptions of technology and their comfort with its use are crucial factors influencing their willingness to use it. As highlighted in the study by Burst et al. [27], participants in our studies shared similar perspectives when assessing different wearable device options. The appearance and interaction modalities of these devices play a significant role in shaping users' experiences and their overall willingness to use them into their daily lives.

Kottl et al. [22] identified age as a key barrier that hinders the design, adoption, and use of digital technologies. In line with our findings, participants expressed a strong preference for jewelry-inspired designs, appreciating their dual functionality—combining portability with aesthetic appeal. These designs seamlessly integrate into daily routines, offering a familiar and discreet form factor that aligns with users' lifestyle needs. Notably, the preference for such designs underscores the importance of addressing the social stigma surrounding menopause. By avoiding overt signals of aging or health conditions, jewelry-inspired designs have the potential as a non-stigmatizing and non-labeling approach. This finding highlights the critical role of inconspicuous design in promoting user acceptance and sustained engagement, ultimately supporting the emotional well-being of individuals navigating menopause.

Moreover, as discussed in Section 4.2.3, the interaction modality plays a crucial role in shaping user experiences. Participants expressed concerns that the petal-like motions incorporated into the design could potentially heighten their anxiety and discomfort rather than providing a calming effect. Their feedback emphasized the need for designers to carefully consider interaction modalities to ensure they do not inadvertently introduce additional negative emotions or stress.

6 Conclusion

Overall, this study explored the feasibility of wearable devices for emotional support during menopause through two workshops. "Calmora" as a design probe effectively encouraged participants to share their experiences and insights. The study concludes that incorporating natural design elements can effectively support the management of negative emotions. In terms of digital technology and wearable design, the focus should be on emotional guidance and management. Additionally, the appearance should avoid emphasizing age or menopause, as this could create psychological barriers to usage.

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A Appendix

Table 1: Participant Information of Study 1 workshop

Participants	Age	Residential situation and Daily responsibilities	Menopause Stage
P1	50	Living with spouse and children; Freelance work	menopause
P2	62	Living with spouse and brother and sister-in-law; Need to care for a spouse in poor health; Retired	post-menopause
P3	50	Three generations living together; Need to care for adolescent children; Full-time work	pre-menopause
P4	50	Taking care of a child alone; Menopause is the child's adolescence; Full-time work	post-menopause
P5	46	Living with spouse and children; Full-time work	pre-menopause
P6	54	Living with spouse and children; Suffering from lung disease; Need to take care of her mother; Full-time work	menopause
P7	62	Living with spouse; Need to take care of her mother; Retired	post-menopause
P8	66	Three generations living together; Need to take care of grandchildren in primary school; Retired	post-menopause

Table 2: Participant Information of Study 2 Focus Group

Participants	Age	Length of Menopause	Technology Usage	Current Strategies
F1	49	2 years	Online posts and educational videos related to menopause; Long-term use of iWatch	Exercises; Medications; Social interaction
F2	54	4 years	Online posts and education videos related to menopause	Exercises; Medications; Social interaction; Psychological therapies
F3	55	2 years	Online posts and education videos related to menopause; Exercise app	Exercises; Social interaction
F4	54	4 years	Online posts and education videos related to menopause	Social interaction
F5	53	8 years	Online posts and education videos related to menopause	Exercises; Medications; Social interaction
F6	54	3 years	N/A	Exercises; Medications